

Counting Fruits with State-of-the-Art Multi-View-Stereo Approaches

Bachelor Thesis in Computer Science

presented by

Alexander Link

born on 29.08.2004 in Bamberg

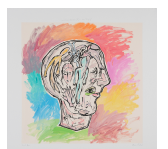
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Abstract

A brief and concise summary of the thesis containing the problem statement and how it was solved in about five sentences. After having read the abstract, a reader must know what the thesis is about.

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Chapter 1

Introduction

The introduction needs to answer the following questions:

- Which area of visual computing are we in? (scientific visualization, appearance modeling, Monte Carlo rendering, etc.)
- Which problem are we going to solve?
- Why is this a challenging problem? What makes it difficult?
- Who will be the user of this method? Who benefits from it?
- How are you going to solve the problem?
- How will you evaluate the method?
- What are your most important findings?

The automatic counting of fruit from images is a key task in precision agriculture, where accurate yield estimates support harvest planning and resource allocation. Reconstructing a fruit tree scene in 3D from a set of RGB images and counting the fruit it contains is appealing because it requires only inexpensive and widely available camera hardware, rather than specialized depth sensors.

citation

Recent Work such as FruitNeRF [**meyer2024fruitnerf**] show that this is possible by training a Neural Radiance Field per scene and counting the fruits in the resulting volumetric representation. This approach is effective but requires a time-consuming scene-specific optimization. In this work, we instead investigate whether a feed-forward multi-view stereo model, VGGT [**wang2025vggt**], can replace scene-by-scene training. Instead of optimizing a representation of each scene, VGGT predicts a 3D reconstruction in a single forward pass. This promises a significant reduction in runtime, but raises the central question of whether the resulting geometry is sufficiently accurate for reliable counting.

Do I have to cite this?

Counting fruits in this scenario is challenging for several reasons: Fruits are often obscured by foliage and each other, the appearance of a scene varies with changing lighting, and the geometric complexity of a tree canopy makes a consistent reconstruction difficult. To reduce occlusion and the risk of missing apples, a scene should ideally be captured from all sides, but the number of views is limited by hardware constraints such as available memory. The cameras are therefore sparsely distributed across a full 360-degree range, and widely spaced views often capture completely different regions of the tree. This

makes it difficult to establish correspondences between views, so that individual seen only by a few cameras are at risk of being fragmented in the reconstruction.

This results in two types of opposing types of errors. Individual apples observed by only a few cameras yield unstable correspondences and are easily split into multiple detections, while dense clusters of apples seen by many cameras may merge into a single detection. A single set of static parameters can therefore rarely resolve both cases optimally at the same time.

Our approach maps semantic information in the form of 2D instance masks, obtained using SAM3 [sam3], across views via the 3D representation predicted by VGGT. The fruits are then counted by combining a clustering step with a preceding partial instance association that supports the clustering.

We evaluate the approach in different ways on multiple datasets

Our key findings suggest that the scene complexity is the decisive factor for counting accuracy, since the quality of the VGGT reconstruction varies significantly depending on lighting, depth complexity, and whether the input is synthetic or real.

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Chapter 2

Related Work

2.1 Preparation of the Bibliography

To add references, first add the corresponding BibTeX entries in the file `bibliography.bib`.

Journal papers use the `@article` entry, which needs at least:

- **title:** Title of the paper
- **author:** Separate authors with "and" and use comma to separate last from first names
- **journal:** Name of the journal.
- **volume:** Volume number. Each year a new volume is created.
- **number:** Issue number within a year. The journal might have multiple issues in a year.
- **pages:** Page range.
- **year:** Year of appearance.

Conference papers use the `@inproceedings` entry, which needs at least:

- **title:** Title of the paper.
- **author:** Separate authors with "and" and use comma to separate last from first names
- **booktitle:** Name of the conference.
- **pages:** Page range.
- **year:** Year of appearance.

2.2 Citing a Paper

To cite a paper in the LaTeX document, use the `cite` command. You can use indirect quotations at the end of a sentence [6].

Or you directly reference an author, such as: Weyrich et al. [6] described a method. If a paper has more than two authors, it is common to use “et al.”, which means “and colleagues”.

There are also several commands that can help when citing a paper, e.g., when you want to cite in a given paper in a text: Weyrich et al. [6].

`cite` Cite only square brackets: [6]

`cite` Cite multiple sources: [2, 4]

`textcite` Cite textual: Egger et al. [1]

`textcite` Cite multiple sources: Weyrich et al. [6] and Harrer et al. [3]

`citeyear` Cite the year: 2023.

Chapter 3

Method

3.1 Tense

Mind your use of present and past tense. Reports are typically written in present tense, except for the related work and conclusion section, which are usually in past tense. If you describe what past authors did, use past tense. If you describe what an algorithm does, which still works the same way today, use present tense.

3.2 I, We, Passive voice

It is very uncommon to write “I”. Instead, either avoid using personal pronouns by switching to a passive voice, or use “we” instead. When reading the word “we”, you can think of this as “the reader and me”.

3.3 Equations

For equations, the `align` command is recommended, since it allows you to align equations at a specific location, such as at the `=` sign.

$$y = f(x) \tag{3.1}$$

$$\boldsymbol{f} = \boldsymbol{m} \cdot \boldsymbol{a}^\top \tag{3.2}$$

Equations are referenced using `eqref`, for example Equation (3.1) and Equation (3.2). See `macros.tex` for a range of convenient macros, e.g., vector $\boldsymbol{x}$ and matrix $\boldsymbol{M}$ symbols

3.4 Figures

Figure placement is best done with the default `figure` environment. To reference a figure use `figref`, for example Figure 3.1. If possible, always create figures as vector graphics. For example an overview / pipeline graphic of your method should always be a vector graphic. Vector graphics allow zooming into figures without a loss in quality of the figure. Vector graphics can be exported in PDF format from common editing programs, which allows including them in Latex with `includegraphics`. Raster

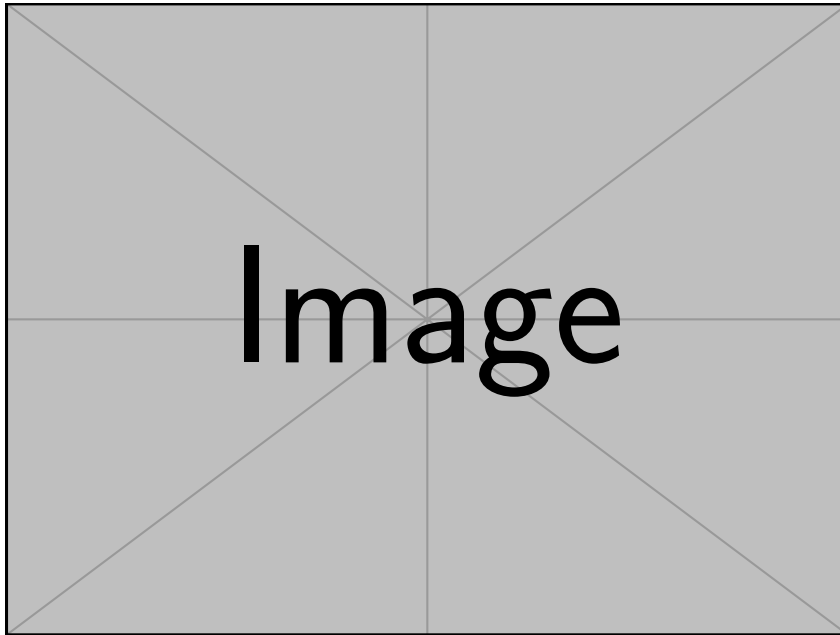


Figure 3.1: Caption of the figure. Always cite the source of the image in the caption, unless you created the image yourself.

	Metric 1↑	Metric 2↑	Metric 3↓	Metric 4↓
Method 1	23.1388	0.8611	0.2479	0.1846
Method 2	28.3813	0.9353	0.1831	0.1047
Ours	26.2468	0.9257	0.1509	0.0915

Table 3.1: Caption of the Table

images should only be used in your thesis when an image is shown, that is naturally rendered at a given resolution, e.g., a result image of a rendering algorithm.

3.5 Tables

Table placement is best done with the default `table` environment. To reference a table use `tblref`, for example Table 3.1.

Chapter 4

Implementation

While the previous section aims to describe your method in a platform independent way, this section presents all the details to reproduce your work.

4.1 Listings

Not all theses require the inclusion of source code. If it is necessary, Code placement is best done with the `lstlisting` environment.

```
1 # Comment
2 def function(parameter):
3     print("This is an example Python listing")
4     return True
```

Listing 4.1: Caption of the Python Listing

```
1 // Comment
2 bool function(int parameter){
3     std::cout << "This is an example C++ listing" << std::endl;
4     return true;
5 }
```

Listing 4.2: Caption of the C++ Listing

To reference a listing use `lstref`, for example Listing 4.1.

Chapter 5

Results

Describe your qualitative and quantitative evaluations and present your results. Discuss the limitations of your work.

Chapter 6

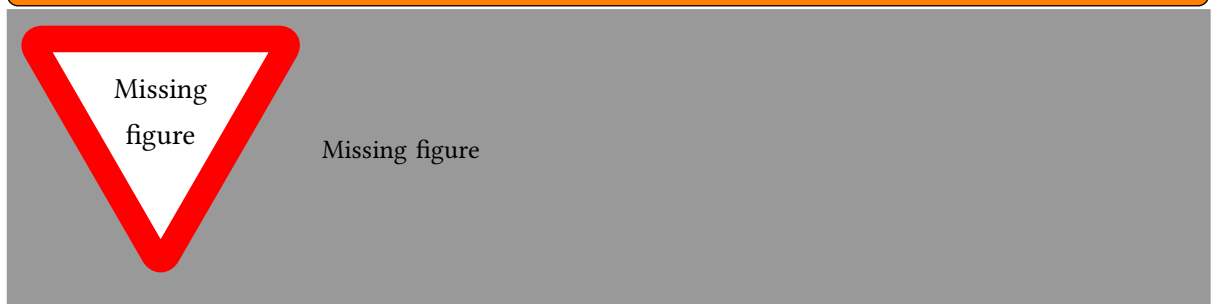
Conclusions

Give a brief summary of your work and your contributions. Afterwards, briefly outline potential opportunities for future work.

6.1 Tips

As simple package to keep track of unfinished areas of the thesis is the `todonotes` package.

Inline TODO



This is
a simple
TODO
note

Create a list of remaining TODOs with `listoftodos`. Don't forget to remove the `listoftodos` before submitting your thesis.

Todo list

citation	1
Do I have to cite this?	1
datasets, performance	2
key findings	2
This is a simple TODO note	11
Inline TODO	11
Figure: Missing figure	11

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LIST OF LISTINGS

Bibliography

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- [6] Tim Weyrich, Wojciech Matusik, Hanspeter Pfister, Bernd Bickel, Craig Donner, Chien Tu, Janet McAndless, Jinho Lee, Addy Ngan, Henrik Wann Jensen, et al. “Analysis of human faces using a measurement-based skin reflectance model”. In: *ACM Transactions on Graphics (ToG)* 25.3 (2006), pp. 1013–1024.

BIBLIOGRAPHY

Declaration

I certify that I have written this thesis without outside help and without using sources other than those indicated, and that the thesis has not been submitted in the same or similar form to any other examination authority and has not been accepted by them as part of an examination. All statements that have been taken over verbatim or in spirit are marked as such.

During the preparation of this work, AI technologies were used to assist in the writing process. Specifically, <AI Tool 1 (Company, City, State, Country)> was used in order to check for grammar and style consistency, and <AI Tool 2 (Company, City, State, Country)> was used in order to assist with rephrasing and improving readability. After using these tools, the manuscript was carefully reviewed and the content was edited as needed. No tools or services were used for content generation. The content of this thesis was entirely created by myself, and I ensured that all material presented reflects my original work and research. I take full responsibility for the content of the thesis.

I agree that the thesis may be published and that it may be referred to in scientific publications.

The Friedrich-Alexander-Universität Erlangen-Nürnberg, represented by the Professorship for Cognitive Computer Vision, is granted a (non-exclusive) right to use this work and the programs created in connection with it.

Erlangen, 01.10.2026

(Alexander Link)